

The effectiveness of quadrupled semitendinosus graft technique in anterior cruciate ligament reconstruction: A network meta-analysis assessing various graft preparation techniques

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Abstract

Purpose: This study aims to perform a network meta-analysis of hamstring graft preparation techniques to enhance anterior cruciate ligament (ACL) reconstruction guidelines and inform clinical decision-making in patients with primary ACL rupture.

Methods: A review of the literature, from 1 January 1990, to 31 August 2023, was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines, focusing on the clinical outcomes of various hamstring graft preparation techniques. Forty-six studies (over 4800 knees) were analysed. Eight graft compositions: doubled hamstring (ST/G)—1978 grafts, doubled hamstring with augmentation (ST/G+A)—586 grafts, tripled semitendinosus (3ST)—124 grafts, quadrupled semitendinosus (4ST)—1273 grafts, five-strand tripled semitendinosus + doubled gracilis (3ST/2GR)—839 grafts, six-strand tripled semitendinosus + tripled gracilis (3ST/3GR)—335 grafts, seven-strand quadrupled semitendinosus + tripled gracilis (4ST/3GR)—11 grafts and ≥eight strands—24 grafts were compared, considering graft sizes, laxity, muscle strength, range-of-motion, patient-reported outcome measures (PROMs), return to sport (RTS) and adverse events.

Results: The 4ST grafts fared better than the ST/G grafts in the International Knee Documentation Committee (IKDC), Knee Injury and Osteoarthritis Outcome Score (KOOS) Sport, KOOS Q and RTS (mean difference [MD], −1.69; $p = 0.0159$; MD, −1.55; $p = 0.0325$; MD, −1.93; $p = 0.001$; odds ratio: 3.13; $p < 0.0001$). The IKDC differed significantly between the 4ST and ST/G+A groups (MD, 1.88; $p = 0.046$). The ST/G+A resulted in the lowest knee laxity, surpassing the ST/G, 3ST and 4ST. The ST/G had the smallest diameter (ST/G vs. ST/G+A: MD, 1.26; 95% confidence interval [CI]: 0.67–1.86, $p < 0.0001$). Reduced failure rates were noted with the 3ST/2GR (3ST/2GR vs. ST/G: MD, 6.93; $p = 0.009$) and 3ST/3GR (3ST/3GR vs. ST/G: MD, 53.64; $p = 0.006$).

Abbreviations: 3ST/2GR, five-strands graft: tripled semitendinosus + doubled gracilis; 3ST/3GR, six-strands graft: tripled semitendinosus + tripled gracilis; 3ST, three-strands graft: tripled semitendinosus; 4ST/3GR, seven-strands graft: quadrupled semitendinosus + tripled gracilis; 4ST, four-strands graft: quadrupled semitendinosus; 8ST/GR, eight-strands graft and more; ACL, anterior cruciate ligament; ACLR, anterior cruciate ligament reconstruction; PROMs, patient-reported outcome measures; RTS, return to sport; ST/G+A, four-strands graft with augmentation: doubled semitendinosus + doubled gracilis with augmentation; ST/G, four-strands graft: doubled semitendinosus + doubled gracilis.

Conclusion: The ideal hamstring graft for ACLR should be individualized. A 4ST graft is likely to yield good PROMs. For high stability and rapid RTS, adding augmentation to the graft is advisable. The ST/G is the thinnest graft possible.

Level of Evidence: Network meta-analysis of level I–III studies.

KEYWORDS

ACL, ACLR, anterior cruciate ligament, graft with augmentation, hamstring graft, quadrupled semitendinosus graft

INTRODUCTION

The doubled semitendinosus and gracilis tendons (ST/G) and quadrupled semitendinosus tendons (4ST) are the most common graft choices in anterior cruciate ligament (ACL) reconstruction (ACLR) [11]. The need for gracilis tendon harvesting is unclear, as numerous reports have indicated use of this tendon does not significantly improve knee stability and affects muscle strength [12, 17, 30, 54]. Moreover, whether sex has any significant impact on treatment outcomes or on the choice of graft is uncertain [30]. ACLR with the hamstrings augmented with allograft or suture tape to provide additional hamstring support (ST/G+A) has also been proposed [1, 42, 64, 67]. Furthermore, when opting to use both hamstring tendons, the configuration most beneficial to the patient is unclear [12, 16, 17, 21, 30, 37, 55, 56].

This study compared different types of hamstrings used in primary ACLR. Patient-reported outcome measures (PROMs); the International Knee Documentation Committee (IKDC), Lysholm, Tegner, Knee Injury and Osteoarthritis Outcome Score [KOOS]), return to sport (RTS) and instability tests, such as the Lachman test, pivot-shift test, anterior-posterior (AP) laxity, AP laxity >3 mm, muscle strength (flexion force, extension force), graft diameter and failure rates were evaluated with a view to providing a guide for selecting a specific hamstring graft.

The study's hypothesis indicates that there is no single optimal hamstring graft type, but by making both direct and indirect comparisons, the most suitable graft could be chosen based on individual patient's needs and expectations.

METHODS

Study selection

Using a common search strategy, three independent reviewers (anonymized) performed the literature search and independently extracted data, in a manner following the 2009 Preferred Reporting Items for

Systematic Reviews and Meta-Analyses (PRISMA) statement [39]. All discrepancies were reviewed by a fourth author (anonymized) and were discussed to consensus amongst the authors. Titles and abstracts of the identified reports were reviewed. Potentially eligible articles were then analysed by full-text review. All analyses were performed with guidance published by Karlsson et al. [46]. The reference lists of the identified investigations were manually searched to identify further reports. The study protocol was registered in PROSPERO under the number: (357171) before starting the study [53].

Search strategy

Using a comprehensive search strategy, the PubMed, EMBASE, clinicaltrials.gov and Cochrane Library databases were systematically searched for papers published from 1 January 1990 to 31 August 2023. The following search algorithm was used: (semitendinosus OR gracilis OR hamstring) AND (ACL OR ACLR or Anterior Cruciate).

Eligibility criteria

Randomized controlled trials (RCT) and non-RCTs comparing different types of hamstring autografts used in ACLR surgery for patients with primary ACL rupture that were published in peer-reviewed journals were included in this network meta-analysis. The inclusion of both RCTs and non-RCTs aimed to facilitate subgroup analysis (RCT vs. non-RCT) and to broaden the evidence base. This analysis included 5170 grafts involving eight different graft compositions: doubled hamstring (ST/G; 1978 grafts), doubled hamstring with augmentation (ST/G+A; 586 grafts), tripled semitendinosus (3ST; 124 grafts), quadrupled semitendinosus (4ST; 1273 grafts), five-strand tripled semitendinosus + doubled gracilis (3ST/2GR; 839 grafts), six-strand tripled semitendinosus + tripled gracilis (3ST/3GR; 335 grafts), seven-strand quadrupled semitendinosus + tripled gracilis (4ST/3GR; 11 grafts)

TABLE 1 Hamstring graft types analysed.

Abbreviation	Graft composition
ST/G	Four strands: doubled semitendinosus + doubled gracilis
ST/G+A	Four strands with augmentation: doubled semitendinosus + doubled gracilis with augmentation
3ST	Three strands: tripled semitendinosus
4ST	Four strands: quadrupled semitendinosus
3ST/2GR	Five strands: tripled semitendinosus + doubled gracilis
3ST/3GR	Six strands: tripled semitendinosus + tripled gracilis
4ST/3GR	Seven strands: quadrupled semitendinosus + tripled gracilis
8ST/GR	Eight strands and more

and >eight-strand grafts (24 grafts) (Table 1). There were no restrictions on language or patient age. Exclusion criteria included studies of ACL grafts other than hamstring (bone to bone graft, quadriceps femoris graft, etc.), concomitant surgical procedures (osteotomies, multiligament reconstruction) and animal or cadaveric studies.

Data extraction

The following data were extracted from each paper by two independent reviewers using a predetermined data sheet: authors, year of publication, type of intervention, details of interventions used in the study, sample size (randomized and analysed), outcome measurements, follow-up period, main results, failures, percentage and type of adverse events included in the publication and level of evidence.

Level of evidence and risk of bias assessment

The Oxford Centre for Evidence-Based Medicine Levels of Evidence Working Group criteria were used to evaluate the level of evidence. The Cochrane Collaboration Risk-of-bias revised tool (RoB 2) was used to assess the risk-of-bias and methodological quality of evidence of RCTs [38, 58]. The following domains were evaluated and assessed for risk of bias (low, unclear or high): bias domain and signalling question, bias due to deviations from intended interventions, bias due to missing outcome data, bias in the measurement of the outcome, bias in selection of the reported result and

overall bias. Nonrandomized trials were assessed using the ROBINS-I tool for bias. The assessment covered biases due to confounding, in participant selection, in intervention classification, due to deviations from intended interventions, due to missing data, in the measurement of outcomes and in the selection of reported results [20]. Overall bias was categorized as of low, moderate, high or critical risk. Sensitivity analyses were performed to evaluate the impact of biased or large studies on model fit, relative effects and treatment rankings.

Outcome measures

Outcome measures collected included PROMs (IKDC, Lysholm score, Tegner activity scale, KOOS), RTS and instability tests, such as the Lachman test, pivot-shift test, AP laxity and AP laxity >3 mm. Additionally, muscle strength (flexion force, extension force), graft diameter and failures were measured. All collected data were analysed with respect to the following interventions described in Table 1.

Statistical analysis

Qualitative statistical analysis and network meta-analysis were performed using R™ software 4.3.1 (R Foundation for Statistical Computing) [47]. To obtain a single pooled effect from a large number of underpowered studies, both RCTs and non-RCTs were analysed, even though this increased heterogeneity. A frequentist approach to network meta-analysis with a random-effects or common-effect model (for studies with below 25% I^2 and nonsignificant heterogeneity) was used, employing the netmeta package version 2.8-2. Effect sizes were reported as mean differences (MD, for continuous outcomes) or odds ratios (OR, for dichotomous outcomes) with 95% confidence intervals (CIs). The league tables and network plots were generated with the dplyr package version 1.1.2. Publication bias was analysed using a comparison-adjusted Funnel plot, Egger's multivariate test method and Rosenthal's file drawer method, as well as relative risk (RR) for Publication Bias Calculation with the meta and metofo packages, versions 6.5.0 and 4.4.0, respectively.

RESULTS

The initial literature search identified 11,845 potential articles. Of these, 7471 were extracted after screening the titles and abstracts. The full text of 348 studies was retained for further screening according to the inclusion and exclusion criteria. Finally, 46 articles met

the inclusion criteria and were analysed in this network meta-analysis (Figure 1).

Study characteristics

Overall, 4863 patients were included in 46 studies. The number of studies compared alongside every subgroup and the main characteristics of each article are presented in Table 2.

In terms of the outcomes of interest, IKDC scores were reported in 25 studies, Lysholm scores in 25, Tegner activity scale scores in 14, KOOS scores in 11, RTS in eight, Lachman test results in 12, pivot-shift test results in 17, overall AP laxity in 19, AP laxity >3 mm in

14, flexion force in 17, extension force in 15, graft diameter in 24 and failures in 21 studies. Each of these outcomes was analysed using a forest plot diagram and network geometry, comparing repair using a double hamstring approach to the other subgroups. All correlations are presented in league tables (Supporting Information S1: Tables 1–17). No potential conflicts of interest were recorded amongst the authors of the studies included in the analysis.

PROMs

Thirty-two studies were included concerning patient-reported outcomes. Network comparison showed

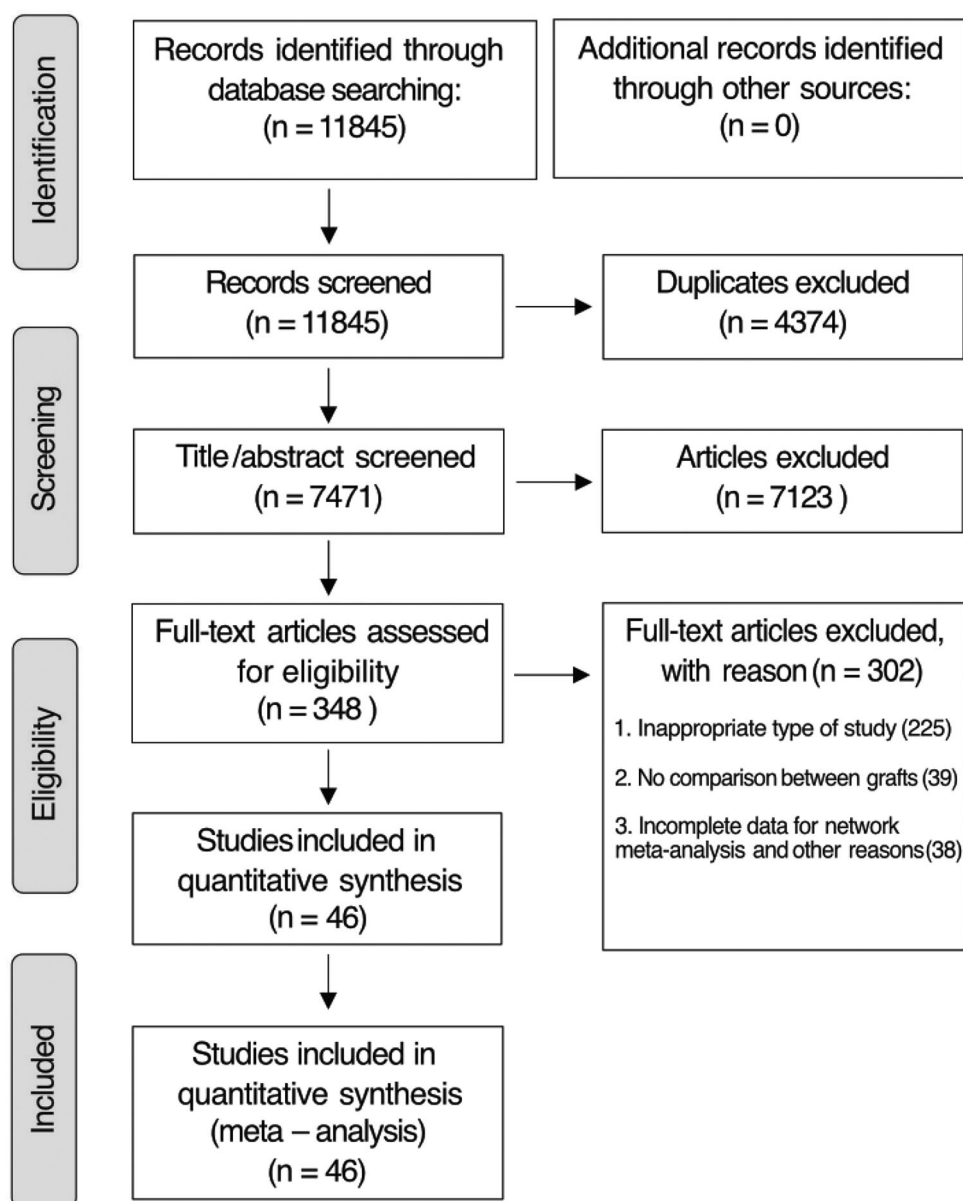


FIGURE 1 Summary of the search process following the PRISMA guidelines. PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

TABLE 2 Main characteristics of the studies included in meta-analysis.

No	Study ID	Study type	Level of evidence	Study period	Country	Type of tendon graft	Number of patients	Patient age (years)	Sex (male%)	Time to surgery	Follow-up period	Outcomes
1	Adams et al. [1]	Retrospective cohort study	III	01.2007–12.2017	USA	ST/G versus ST/G+A	112	29.2 and 28.8	85	Not mentioned	65.3 months	Failure rates, KOOS
2	Alkhalaf et al. [2]	Case series	IV	2012–2018	Kuwait	3–10 stands ST/G versus ST/G+A	711	28.72	Not mentioned	Not mentioned	2 years	Graft diameters, failure rates
3	Ardern et al. [3]	Retrospective cohort study	III	06.2004–06.2007	Australia	4ST versus ST/G	50	26.1 ± 6.9 and 24.4 ± 4.9	70	Not mentioned	31.4 months and 33.5 months	Active and passive knee range-of-motion and anterior knee laxity (KT-1000); IKDC, Isokinetic hamstring strength and isometric hamstring strength
4	Attia et al. [4]	Cohort study	III	01.2011–11.2017	Qatar	4ST versus 3ST2GR versus 3ST/3GR	413	31.54	95	Not mentioned	44.27 months	Failure rates
5	Aujla et al. [5]	Comparative cohort study	III	04.2014–12.2017	Australia	ST/G versus ST/G+A	196	26.80	67	14.8 ± 21.9 weeks 14.2 ± 31.2 weeks	2 years	IKDC, KOOS, Lysholm, Tegner and the NSARS; measures: knee laxity, maximal isokinetic knee flexion and extension strength, four functional hop tests
6	Berenius et al. [7]	Retrospective cohort study	III	Not mentioned	Australia	4ST versus ST/G	20	26 (9) and 26 (7)	50	36 (27–45) months	37 (6) and 36 (4) months	IKDC, RTS, knee range-of-motion, Lachman, pivot-shift, KT-1000; test of hamstring strength, tibial rotation
7	Burrus et al. [9]	Retrospective comparative study	III	01.2007–12.2014	USA	ST/G versus ST/G+A	58	26.6 ± 10.3 and 27.2 ± 9.6	34	3.7 ± 5.6 and 5.2 ± 6.9 months	44.4 ± 16.9 months and 48.0 ± 15.2 months	Failure rates, Lysholm, IKDC
8	Calvo et al. [10]	Retrospective comparative study.	III	01.2012–12.2013	Chile	4ST versus 3ST/2GR	152	13–53	Not mentioned	<3 months	30 months	Reoperation rate, graft diameter, Lysholm, IKDC, anterior laxity measured with a KT-1000

(Continues)

TABLE 2 (Continued)

No	Study ID	Study type	Level of evidence	Study period	Country	Type of tendon graft	Number of patients	Patient age (years)	Sex (male%)	Time to surgery	Follow-up period	Outcomes
9	De Geofroy et al. [13]	Single centre retrospective comparative study	III	2012–2019	France	4ST versus ST/G	133	28.5 ± 6.23 and 29.8 ± 6.46	97	221 ± 437 and 280 ± 336 days	4 years	Return to duty, isokinetic results, sports results (Cooper, walk-run); X-ray analysis, complications and failure rates
10	Demirag et al. [14]	Prospective cohort study	III	11.2005–03.2008	Turkey	ST/G versus ST/G+A	40	30 (19–40)	90	Not mentioned	24.3 months	IKDC, Lysholm, instability tests, patient satisfaction questionnaires and incidences of cyclops lesions and arthrofibrosis; tibial and femoral tunnel widening
11	Drocco et al. [16]	Case-control study	III	2012–2014	Italy	3ST versus ST/G	90	29.5 ± 10.0, 27.7 ± 9.0	44	Not mentioned	12–30 months	IKDC, Tegner, Lysholm, in antero-posterior laxity (Arthrometer KT-1000), the one-leg-hop test, isokinetic test, RTS
12	Gobbi et al. [17]	Randomized controlled trial	I	01.1996–12.2001	Italy	4ST versus ST/G	97	28, 80	57	Not mentioned	36 months (24–52)	Lysholm, Tegner, IKDC, clinical examination, laxity, isokinetic testing and single-leg hop and jump tests
13	Heyworth et al. [19]	Cohort study	III	01.2003–12.2017	USA	ST/G versus ST/G+A	111	15.8 ± 1.9 and 16.1 ± 1.5	20	2.2 [1.2–3.1] and 1.4 [1.1–2.1] months	2.5 years	IKDC, Lysholm, Tegner, pivot-shift test, Lachman test failure rates
14	Hu et al. [21]	Retrospective comparative study	III	01.2009–10.2016	USA	4ST versus ST/G	84	18–40	49	Not mentioned	3.09 ± 0.31 and 3.26 ± 0.23 years	Examination (effusion, joint line tenderness, McMurray, pivot-shift, range-of-motion); laxity test (KT-1000), strength test, functional tests (single-leg-hop test, single-leg-squat test); IKDC, Lysholm, KOOS S
15	Ignazzi et al. [22]	Retrospective cohort study	III	2013–2018	USA	4ST versus 3ST/2GR	74	31.8 ± 13.5/ 23.6 ± 11.0	54	Not mentioned	3.0 ± 1.5 years and 3.3 ± 1.3 years	graft diameter, revision rates, RTS
16	Jimenez et al. [23]	Research Article	III	2010–2016	USA	ST/G versus ST/G+A	46	31 (17–51) and 33.3 (15–48)	59	Not mentioned	31.5 months (range 12–63 months)	Failure rates, IKDC, KOOS, RTS

TABLE 2 (Continued)

No	Study ID	Study type	Level of evidence	Study period	Country	Type of tendon graft	Number of patients	Patient age (years)	Sex (male%)	Time to surgery	Follow-up period	Outcomes
17	Karimi-Mobarakeh et al. [24]	Randomized controlled trial	I	03.2011–03.2012	Iran	4ST versus ST/G	119	29.9 ± 7.8 and 32.4 ± 6.3	82.4	At least 1.5 months	2.41 ± 0.31 years in the ST/G group compared to 3.26 ± 0.23 years	Surgical complications: IKDC; KOOS; Lysholm; strength of the knee isometric flexors; and flexion and extension loss; laxity
18	Kehribar et al. [25]	Retrospective cohort study										
III	April 2019–March 2021	Turkey	ST/G versus 4ST	39	18–35	not mentioned	6 months	Lysholm, Tegner, IKDC, single-leg-hop test				
19	Kouloumentas et al. [27]	Randomized Clinical Study	I	2015–2016	Greece	4ST versus ST/G	90	28.7 ± 11.3	61	Not mentioned	24 months	IKDC, KOOS, KSS, Anterior tibial translation, isokinetic tests
20	Krishna et al. [28]	Cohort study	II	12.2014–06.2016	Singapore	ST/G versus 3ST/2GR	45	25.6 ± 7.3 and 29.2 ± 8.0	79	Not mentioned	At least 1 year	Lysholm, KOOS symptoms, KOOS pain, KOOS ADLs, KOOS Sports, KOOS quality of life and SF-36 PCS
21	Królikowska et al. [29]	Retrospective comparative study	III		Poland	4ST versus ST/G	40	27.00	100	3–6 months	24 weeks	Isometric torque
22	Kulinski et al. [30]	Randomized controlled trial	I	2015–2017	Poland	4ST versus ST/G	140	31 (18–45)	78.5	8.1 (1.5–209.0) and 7.0 (1.5–67.4) months	4.5 years	VAS, IKDC, KOOS, Tegner, Lysholm, ATT (KT-1000)
23	Kumar et al. [31]	Randomized controlled trial	I	Not mentioned	India	4ST versus ST/G	60	28.50	93	7.6 months and 6.46 months	Not mentioned	Anterior drawer test, Lachman, pivot-shift, mean anterior tibial translation measured in 90° of knee flexion, Lysholm
24	Kyung et al. [32]	Retrospective comparative study	IV	2004–2010	Korea	4ST versus ST/G	144	28.0 ± 10.6 and 31.3 ± 12.3	100	5.8 ± 8.1 and 6.0 ± 7.5 months	26.4 and 27.5 months	Lachman, pivot-shift and one-leg hop tests; an isokinetic test; KT-2000 arthrometer; assessment of return to preinjury activities; Lysholm, Tegner, IKDC

(Continues)

TABLE 2 (Continued)

No	Study ID	Study type	Level of evidence	Study period	Country	Type of tendon graft	Number of patients	Patient age (years)	Sex (male%)	Time to surgery	Follow-up period	Outcomes
25	Lee et al. [33]	Retrospective comparative study	III	06.2009–03.2014	Korea	4ST versus ST/G	120	27.4 ± 6.6 and 26.9 ± 7.3	93	2.7 ± 1.7 months and 2.9 ± 1.5 months	37.5 ± 15.9 and 36.8 ± 6.1 months	Lysholm, IKDC SKF, Postoperative Tegner, side-to-side difference on KT-2000 (mm), graft diameter
26	Mahiroglu et al. [34]	Prospective comparative study										
III	2007–2009	Turkey	4ST versus ST/G	98	31.14 ± 7.80 and 28.69 ± 8.07	Not mentioned	26.04 ± 2.53 and 25.93 ± 2.66	Lysholm, Tegner, IKDC				
27	Manchado et al. [35]	Retrospective observational study	III	2013	Spain	4ST versus ST/G	44	26.8 ± 8.5 and 25.8 ± 8.9	100	Not mentioned	5 years	STSD, Lysholm, IKDC, LSI, HI/Q ratio strength
28	Nazari et al. [37]	Prospective cohort study	III	07.2018–03.2020	United Arab Emirates	3ST, 2ST/G, 3ST/2GR, 3ST/3GR	78	33.7 ± 5.9	100	Not mentioned	Not mentioned	Preoperative anthropometric data, graft diameter
29	Parkes et al. [40]	Retrospective comparative study	III	07.2011–07.2017	USA	ST/G versus ST/G+A	108	25.3 (13–44); 24.9 (13–54)	69	7.6 ± 16.0 (0.2–76.2); 7.1 ± 18.6 (0.3–124.4) months	2 years	RTS, range-of-motion, Tegner, Lysholm, IKDC, Lachman, pivot-shift, presence of effusion
30	Pennock et al. [41]	Cohort study	III	2003–2014	USA	ST/G versus ST/G+A	50	15.7 (12–18)	38	Not mentioned	2 years	Failure rates, need for second surgery, contralateral ACL tear, SANE score, patient satisfaction, VAS, Tegner, Lysholm, RTS
31	Perkins et al. [42]	Cohort study	III	08.2012–04.2016	USA	ST/G, 3ST/2GR, 3ST/3GR, ST/G+A	354	15.3 (10–19)	44	Not mentioned	26 months	Graft diameter, failure rates
32	Plaweski et al. [44]	Cohort study	III	03.2012–07.2013	France	4ST versus ST/G	87	25.2 (18–45)	55	6.6 ± 4 and 5.8 ± 4 months	3.2 ± 0.2 years	IKDC, KT-1000 arthrometry, radiologic evaluation of the differential laxity, Tegner, KOOS

TABLE 2 (Continued)

No	Study ID	Study type	Level of evidence	Study period	Country	Type of tendon graft	Number of patients	Patient age (years)	Sex (male%)	Time to surgery	Follow-up period	Outcomes
33	Powell et al. [45]	Retrospective comparative study	III	2017–2021	USA	4ST versus ST/G and ST/G+A	191	18 (17–31.5) and 26 (18–38.25)	48, 35	Not mentioned	6 (3–7) and 4 (2–6) months	Graft diameter
34	Roger et al. [48]	Randomized controlled trial	I	08.2017–08.2018	France	4ST versus ST/G	60	30.5 ± 8.9 and 30.3 ± 8.5	82	1–1/2 months	2 years	IKDC, isokinetic muscle strength recovery (side-to-side differential laxity, balance, loss of flexion/extension), return to work, pain during physical activities, surgical complications
35	Ruffilli et al. [49]	Retrospective comparative study	III	1998–2000	Italy	4ST versus ST/G	51	29.2 ± 3.8	92	Not mentioned	12.1 ± 1.6 years	IKDC, KOOS, Tegner, arthrometric analysis (KT-1000), radiographic evaluation
36	Segawa et al. [54]	Prospective cohort	II	12.1995–12.1998	Japan	3ST versus 4ST versus ST/G	62	20, 80	55	10.2 months	12 months	The peak torque of knee extension and flexion, Lysholm, anteroposterior stability (KT-1000), anteroposterior and lateral radiographs of the knee in maximal extension
37	Sideris et al. [57]	Cohort study	II	12 months	Australia	ST/G versus 3ST/2GR	44	30.0 ± 8.4 and 25.1 ± 4.1	63	16.0 ± 27.8 versus 5.7 ± 3.9 weeks	12 weeks	Graft diameter, flexion/extension in the gait cycle, adduction/abduction in the gait cycle, axial rotation in the gait cycle, relationships between graft size and rotational stability
38	Tashiro et al. [59]	Prospective randomized study	I	03.1996–11.1996	Japan	4ST versus ST/G	85	24.5 ± 7.7 and 24.8 ± 6.4	57	Not mentioned	24 months	IKDC, knee stability (KT-1000), activity level, bilateral thigh muscle strength (Cybex II dynamometer), quadriceps muscle strength, hamstring muscle strength (isokinetic peak torque measurements)
39	Tutkus et al. [61]	Case-control study	III	2014–2015	Lithuania	3ST/3GR versus 3ST/2GR	122	31.57 ± 8.43, 30.29 ± 7.80, 32.09 ± 9.55, 31.14 ± 10.89	74	Not mentioned	Not mentioned	Preoperative anthropometric data, graft diameter

(Continues)

TABLE 2 (Continued)

No	Study ID	Study type	Level of evidence	Study period	Country	Type of tendon graft	Number of patients	Patient age (years)	Sex (male%)	Time to surgery	Follow-up period	Outcomes
40	Walczak et al. [62]	Prospective paired cohort study	III	Not mentioned	USA	ST/G versus 3ST/2GR	46	28, 90	56.5	Not mentioned	Not mentioned	Graft diameter
41	Wan et al. [63]	Prospective cohort study	II	01.2019–06.2021	China	ST/G versus 3ST/2GR	45	18–45	66	Not mentioned	17.5 months	Lysholm, KOOS Symptoms, Pain, ADLs, Sports and Quality of Life; Lachmann, pivot-shift
42	Wang et al. [64]	Retrospective comparative study	III	07.2013–07.2014	China	ST/G versus ST/G+A	57	32.8±9.9; 32.6±10.3	33	1.3±0.8; 1.4±0.8 months	3 years	IKDC, pivot-shift, Lachman, KT-1000 side-to-side differences Tegner, Lysholm, failure rates, physical examinations, magnetic resonance imaging, second-look arthroscopy
43	Yosmaoglu et al. [65]	Prospective cohort	II	Not mentioned	Turkey	4ST versus GST/G	46	29±7 and 28±9	Not mentioned	Not mentioned	12 months	Isokinetic torques of knee extensors and knee flexors, eccentric coordination, concentric coordination, anterior tibial laxity deficits
43	Zaid et al. [66]	Prospective study	II	01.2018–12.2019	China	4ST/ST/G+A	158	28.00	70	25 months	3, 6, 12, 24 months	Lysholm, IKDC, KOOS-ADLS, Tegner, ACL-RSI, NSARS, GRC scale, RTS, arthroscopic outcomes, graft signal intensity, postoperative complications or rerupture rates, objective outcomes included KT-1000 to evaluate graft stiffness and knee stability, Lachman, pivot-shift, knee flexion and extension range-of-motion, the four hop tests, concentric isokinetic quadriceps and hamstring strength
45	Zaid et al. [67]	Randomized clinical study	I	01.2014–04.2023	China	ST/G versus ST/G+A (LARS and	120	26.4±3.5 and 26.1±4.0 and 25.8±4.6	69	25.1±3.2 and 24.8±2.9 and 24.5±3.0	8 years	IKDC, KOOS, Lysholm Tegner, ACL-RSI; Knee Outcome Survey ADL Subscale, NSARS; GRC; KT-1000 to evaluate knee

TABLE 2 (Continued)

No	Study ID	Study type	Level of evidence	Study period	Country	Type of tendon graft	Number of patients	Patient age (years)	Sex (male%)	Time to surgery	Follow-up period	Outcomes
						meshwork of LARS						stability and graft stiffness, pivot-shift, Lachman, knee flexion and extension range-of-motion, four-hop test and concentric isokinetic quadriceps and hamstring strength
46	Zhang et al. [68]	Prospective cohort study	III	02.2016–03.2018	China	ST/G versus ST/G+A	60	Not mentioned	Not mentioned	Not mentioned	18 months	Lysholm, Tegner, IKDC; Lachman, pivot-shift

Abbreviations: ACL, anterior cruciate ligament; ADLs, activities of daily living; GCR, Global Change Rating; IKDC, International Knee Documentation Committee; KOOS, Knee Injury and Osteoarthritis Outcome Score; NSARS, Noyes Sports Activity Rating Scale; RSI, return to sports after injury; RTS, return to sport; SANE, Single Assessment Numeric Evaluation; VAS, visual analogue scale.

significant differences in IKDC, KOOS Q and Sport and RTS. No significant differences in Lysholm scores and KOOS Pain, Symptoms or ADLs were observed. Use of the quadrupled semitendinosus (4ST) approach was superior to that of the ST/G approach in terms of IKDC ($p = 0.0159$), KOOS Sport ($p = 0.0325$), KOOS Q ($p = 0.0012$) and RTS ($p < 0.0001$). The 4ST approach demonstrated significantly better IKDC ($p = 0.046$) and RTS ($p < 0.0001$) results than did the ST/G+A approach. However, the ST/G+A approach yielded significantly better Tegner results than did the ST/G ($p = 0.0014$) and 4ST ($p = 0.0002$) approaches. RTS was significantly faster with the ST/G+A than with the ST/G ($p < 0.0001$) approach (Figure 2).

Laxity

Twenty-six studies reported the results of four different knee instability tests: AP laxity, AP laxity > 3 mm, Lachman test and pivot-shift test. The ST/G+A resulted in the least AP laxity. This approach resulted in significantly better AP > 3 mm and Lachman test results than that of the three other main groups: ST/G, 3ST and 4ST. Additionally, it had a significantly lower prevalence of pivot-shift positive knees than did the 4ST ($p < 0.0001$). The ST/G was more stable than the 3ST and 4ST in terms of AP > 3 mm and the Lachman test. Additionally, in the pivot-shift test, ST/G showed fewer pivot-shift episodes than did the 4ST ($p < 0.0001$) (Figure 3).

Flexion and extension force

Muscle strength, measured by extension force, was reported in 10 studies and showed a significant difference in favour of 3ST compared to ST/G ($p = 0.0066$), ST/G+A ($p = 0.0090$) and 4ST ($p = 0.0085$). However, flexion force was better with 4ST than with ST/G ($p < 0.0001$) (Figure 4).

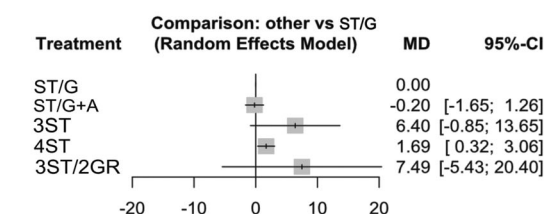
Graft diameter

The smallest graft diameter was assessed in ST/G compared to other groups. Significant differences were found in favour of ST/G+A, 3ST/2GR, 3ST/3GR, 4ST/3GR compared to both groups (ST/G vs. ST/G+A: $p < 0.0001$; ST/G vs. 3ST/2GR: $p = 0.0009$; ST/G vs. 3ST/3GR: $p = 0.0078$; ST/G vs. 4ST/3GR: $p = 0.0442$). Grafts with augment were bigger than 4ST ($p = 0.0139$) (Figure 5).

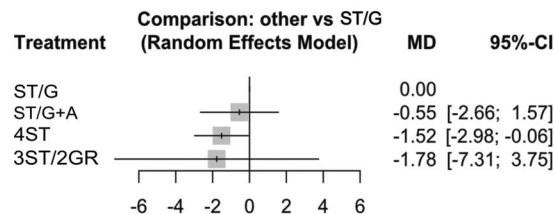
Failures

Interestingly, only the 3ST/2GR and 3ST/3GR groups showed improvements in failures. 3ST/2GR showed

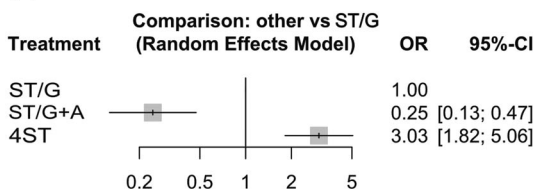
(a) IKDC



(b) KOOS Sport



(c) RTS



(d) Tegner activity score

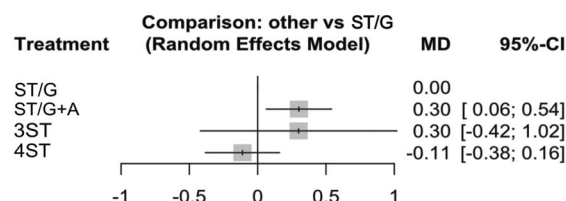
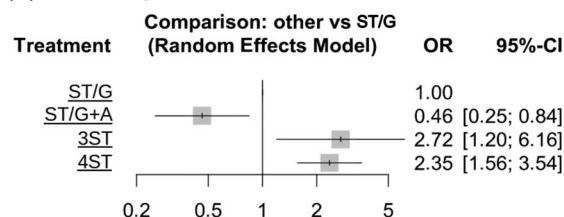


FIGURE 2 Forest plot of International Knee Documentation Committee (IKDC) score (a); Knee Injury and Osteoarthritis Outcome Study (KOOS) Sport (b); return to sport (RTS) (c) and Tegner activity score (d). 95% CI, 95% confidence interval; MD, mean difference; OR, odds ratio.

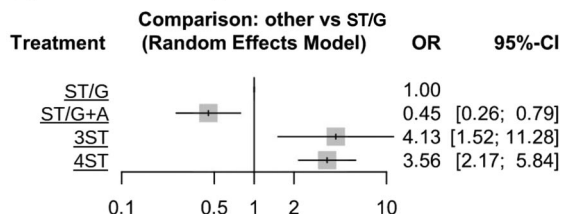
(a) AP laxity (mm)



(b) AP laxity >3mm



(c) Lachmann sign



(d) Pivot-shift

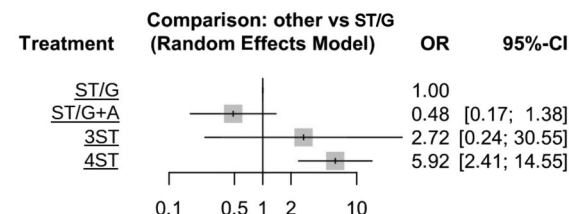
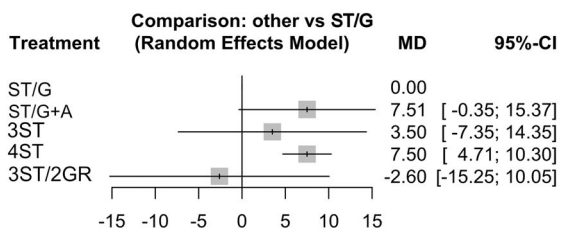


FIGURE 3 Forest plot of AP laxity (mm) (a); AP laxity >3 mm (b); Lachman sign (c) and pivot-shift (d). AP, antero-posterior; 95% CI, 95% confidence interval; OR, odds ratio.

(a) Flexion force



(b) Extension force

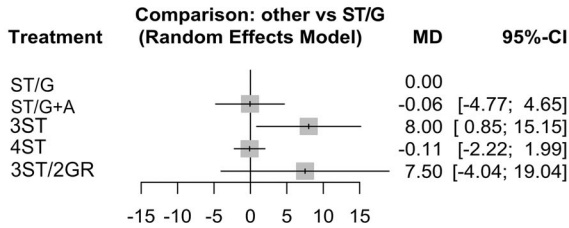


FIGURE 4 Forest plot of flexion force (a) and extension force (b). 95% CI, 95% confidence interval; MD, mean difference.

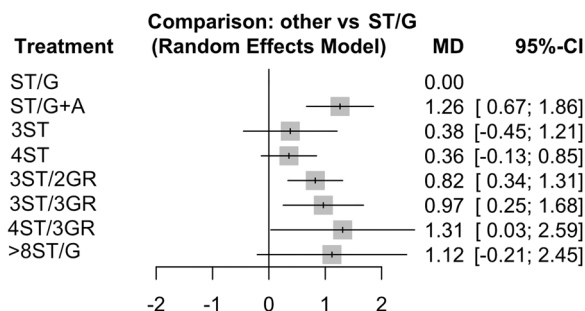


FIGURE 5 Forest plot of graft diameter. 95%-CI, 95% confidence interval; MD, mean difference.

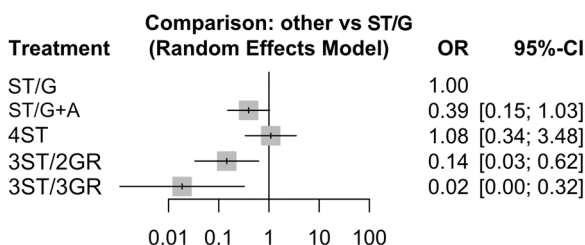


FIGURE 6 Forest plot of failures. 95% CI, 95% confidence interval; OR, odds ratio.

significant improvements as compared to ST/G and 4ST ($p = 0.009$ and $p = 0.005$, respectively). 3ST/3GR also showed significant advantages over ST/G, ST/G +A and 4ST ($p = 0.006$; 3ST/3GR vs. ST/G+A: $p = 0.044$; 3ST/3GR vs. 4ST: $p = 0.007$) in terms of failures (Figure 6).

Risk-of-bias

Of the 46 articles included in this study, eight were randomized trials. They were assessed for risk-of-bias using the ROB2 tool. Two of these articles were categorized as low, four as moderate and two as high risk-of-bias. The remaining 38 articles were non-RCTs. Consequently, they were assessed for bias using the ROBINS-I tool. Ten of these were classified as low, 16 as moderate and 12 as high or very high risk-of-bias. Most of the studies showed potentially high biases in the initial process of randomization, biases due to confounding and unequal treatments in groups, although the final results were improved and presented significantly less high risk-of-bias. The risk-of-bias assessment of all included studies is illustrated in Figure 7 (RCTs) and Figure 8 (non-RCTs).

Sensitivity analysis

Sensitivity analyses were conducted to evaluate the impact of biased or large studies on model fit, relative

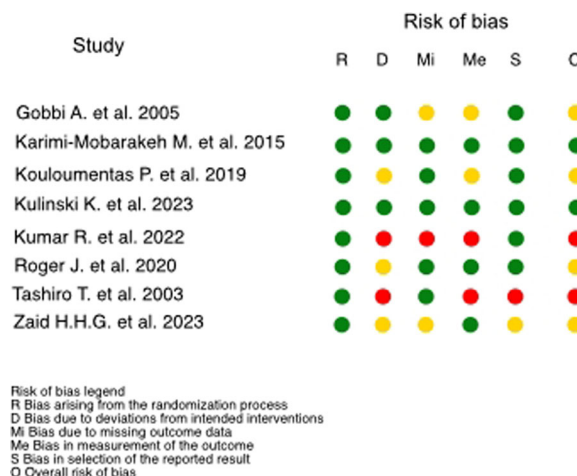


FIGURE 7 Risk-of-bias assessment for the included randomized studies [18, 24, 27, 30, 31, 48, 59, 67].

effects and treatment rankings. To ensure the robustness of the summarized effect sizes, random-effects subgroup analyses were performed for all assessments initially conducted using fixed-effect methods. In the second sensitivity analysis, 14 highly or critically biased studies were excluded from the strength network [2, 4, 5, 14, 16, 31, 32, 41, 59, 61–64, 68]. In the third sensitivity analysis, the largest studies were excluded from the strength network [4, 5, 37, 42, 66]. No significant differences were observed when transforming from fixed-effect to random-effects models. Additionally, upon excluding large or highly biased studies, the results remained grossly unchanged. The estimated failure rate in ST/G+A versus ST/G was significantly lower when large studies were excluded (OR: 0.28; 95% CI: 0.21–0.37, $p = 0.001$). The observed differences in graft diameter upon removal of large or highly biased studies were similar, although nodes with 4ST/3GR and 8ST/GR were removed from the analysis.

The sensitivity analysis supported the robustness of primary findings. The consistency across different models and the negligible impact of excluding large or biased studies indicated that this network meta-analysis provides reliable conclusions.

DISCUSSION

This study indicated that the selection of the appropriate graft for ACLR should be personalized, based on the patient's specific needs, expectations and physical demands. The results of this network meta-analysis confirmed the initial hypothesis that no single, universal, method of folding hamstring grafts is best for all patients undergoing primary ACLR. Nevertheless, the obtained results can be helpful in personalizing and facilitating hamstring graft selection in certain clinical

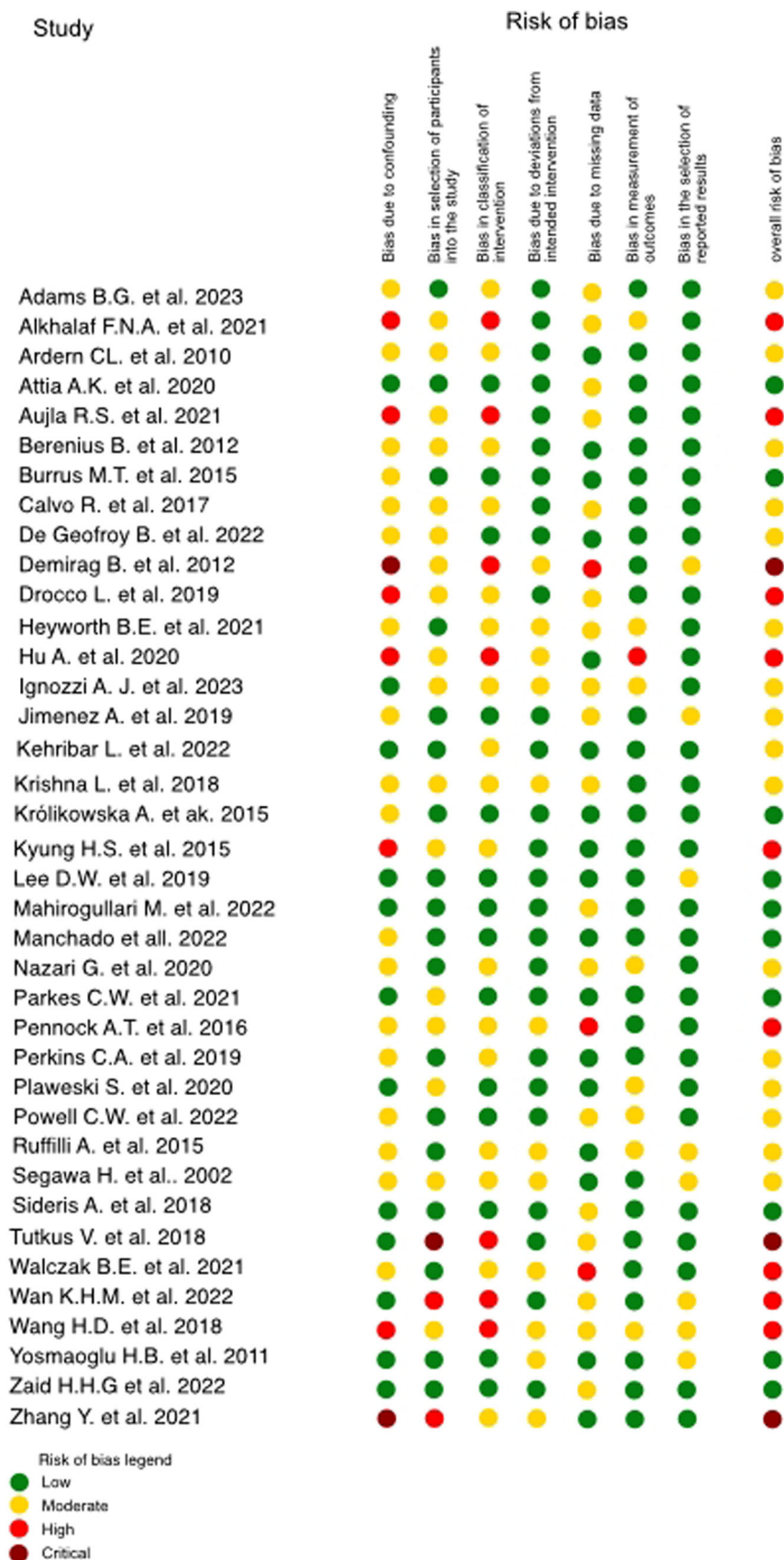


FIGURE 8 Risk-of-bias assessment for the included non-randomized studies [1–5, 7, 9, 10, 13, 14, 16, 19, 21–23, 25, 28, 29, 32–35, 37, 40–42, 44, 45, 49, 54, 57, 61–66, 68].

situations where specific treatment outcomes are desired.

To date, most studies have focused on comparing two specific methods of preparing hamstring grafts. One of the more frequently discussed aspects is the necessity of using the gracilis tendon for ACLR.

One significance found of this study is that doubled hamstring grafts, which are amongst the most commonly chosen hamstring grafts, showed no significant difference in terms of five outcome measures as compared to other hamstring graft preparations. ST/G demonstrated significant advantages as compared to other graft preparations in only four outcomes; more importantly, it showed significant disadvantages in 10 of the outcomes as compared to other groups.

This raises questions that were previously discussed about the need to sacrifice the gracilis tendon. Gobbi et al. found no significant differences between the ST/G and ST groups in terms of knee scores, self-evaluation, clinical findings, knee laxity or functional tests [18]. In a comprehensive analysis, Sharma et al. conducted a systematic review to compare the outcomes of ST grafts and combined ST/G grafts in ACLR. The review revealed no significant differences in PROMs, anterior laxity or isokinetic strength between ST and ST/G groups, particularly at speeds of 180°–240° per second. However, a decrease in hamstring isokinetic strength at 60° per second at flexion angles of 90°, 105° and 110° was noted in the ST/G group [56]. Chin et al. confirmed that, amongst ST and ST/G groups, PROMs and anterior laxity did not differ significantly. However, the ST/G group showed an increased side-to-side deficit in isokinetic peak torque [12].

On the other hand, Kulinski et al. demonstrated that laxity was increased and PROMs were worse amongst females with 4ST than in those with ST/G grafts. Most of the studies included in the present analysis did not provide data on female subgroups [30]. Thus, further well-designed, randomized studies comparing ST/G with ST with a particular focus on differences between the sexes are needed.

The present study expands on these findings by offering a more comprehensive analysis. Unlike previous meta-analyses that compared only ST to ST/G, this study distinguished amongst subtypes of the semitendinosus graft. Given the variability in preparation methods for the semitendinosus graft, such as three- or four-strand configurations, this study provided more detailed outcomes. It revealed statistically significant differences in PROMs and anterior laxity between ST/G and one or multiple groups of ST.

This research supports previous findings that suggested that graft choices should be tailored according to sex, age, activity level and sport [6, 26]. The 4ST presented significantly better PROMs than those of ST/G, but higher laxity than that of ST/G or ST/G+A. In

each of the abovementioned tests, 4ST presented significantly more instances of instability.

Conversely, ST/G+A demonstrated the most stability, but did not show benefits in terms of PROMs, except on the Tegner scale, which rates activity based on work and sport. Here, ST/G+A grafts scored the highest in terms of stability, suggesting better sports performance perceptions in patients. ST/G without augmentation also displayed more stability than did 4ST or 3ST, but fared worse in terms of IKDC, KOOS Sport and Q, and RTS as compared to 4ST, with RTS being significantly faster only than that of ST/G+A. 3ST showed significant improvements only in extension force as compared to ST/G, ST/G+A and 4ST.

These findings suggest that 4ST could well be the preferred option for patients with lower sports expectations who prioritize a pain-free knee for daily activities. ST/G+A might be suitable for active patients for whom returning to high-level sport is crucial. ACLR remains a suitable treatment, particularly for active patients of all age groups, given that an ACL-deficient knee can result in early onset of osteoarthritis, secondary meniscal tears and consequently an increased risk of requiring total knee arthroplasty [15, 43, 52, 60].

The analysis raises many questions. Why do grafts with higher RTS and Tegner scores not perform as well in daily PROMs? Should patients with augmented grafts be allowed to return-to-sport activities faster or would their belief that they can do 'more' increase their risk or is there high heterogeneity within this group?

Other factors, such as graft fixation, graft preparation or graft diameter and length, could have influenced the final outcomes [50, 51]. Graft length was not presented in the included studies, but the ST/G graft is usually longer because, in the ST/G, the ST is only doubled, rather than being folded three, four, five or more times. Length may account for the greater stability of the ST/G and ST/G+A than that of the shorter ST grafts. On the other hand, graft diameter could be related to the better outcome in PROMs. The ST/G is the thinnest graft considered in this meta-analysis and showed inferior outcomes in PROMs. It remains unclear whether differences in PROMs should be attributed to the additional harvest of the gracilis tendon or to the diameter of the graft [8, 36].

This network meta-analysis compared different types of hamstring grafts, which has not been reported previously. The advantage of a network meta-analysis is that it allows for the direct and indirect comparison of treatments using a common comparator and allows for them to be ranked using the *p* value.

However, the study had some limitations. The time frames and length of assessment differed between studies. ACLR surgical technique (graft fixation, tunnel placement, etc.) and postoperative rehabilitation programme may have been sources of heterogeneity. Additionally, the variety of scores, scales and laxity

assessments used in the included studies may have been a reason for high heterogeneity. Comparison of clinical trials with the same follow-up period and measured outcomes may have yielded more consistent findings. The most important outcomes were PROMs and instability assessments. Since the analysed results were derived from studies with mixed cohorts (both female and male), the findings may have been biased due to sex differences in available data. Fourteen studies were at high or critically high risk-of-bias, which could have influenced the outcomes. Too few RCTs included PROMs and outcomes, as illustrated by the network geometries in Figures S1–S4.

Well-designed prospective RCTs incorporating data on anatomical, biomechanical, graft dimensions, length, PROMs, stability, graft preparation and stabilization techniques in both sexes could help to determine the optimal hamstring graft type. This study should prompt clinicians to consider all possible options and consequences when choosing a graft before the ACLR surgery, with a focus on choosing the graft that suits the individual patient best.

CONCLUSION

The ideal choice of a hamstring graft for ACLR remains debated. When selecting a graft, various factors should be considered to ensure the best possible outcome for the patient [50, 51]. According to this network analysis, the 4ST graft offers a higher likelihood of patient satisfaction as measured by some PROMs, making it a strong candidate for those who prioritize overall satisfaction and function. For patients who place a high value on a rapid RTS and knee stability, the ST/G+A graft is advisable. This graft provides the support and strength needed for demanding activities, making it ideal for athletes and highly active individuals. In situations where anatomical or technical conditions necessitate the use of the thinnest possible graft (e.g. children), the ST/G is the recommended option, as this graft can accommodate the need for a slimmer profile without compromising on the effectiveness of the ACLR. Furthermore, it is worth noting that the use of 3ST/2GR or 3ST/3GR grafts statistically resulted in the lowest failure rates. This finding highlighted the reliability and durability of these graft combinations, making them a prudent choice for patients concerned about long-term success and the minimization of reoperation risks. Overall, the selection of the appropriate graft should be personalized, based on the patient's specific needs, expectations and physical demands, to achieve optimal outcomes in ACLR.

AUTHOR CONTRIBUTIONS

Conceptualization: Rafał Kamiński. *Data curation:* Kamila Malesa, Ewa Tramś, Krzysztof Kuliński and

Rafał Kamiński. *Formal analysis:* Rafał Kamiński and Kamila Malesa. *Funding acquisition:* Rafał Kamiński. *Investigation:* Kamila Malesa, Krzysztof Kuliński, Ewa Tramś and Rafał Kamiński. *Methodology:* Rafał Kamiński and Ewa Tramś. *Project administration:* Ewa Tramś and Kamila Malesa. *Resources:* Rafał Kamiński. *Software:* Rafał Kamiński. *Supervision:* Rafał Kamiński. *Validation:* Rafał Kamiński and Krzysztof Kuliński. *Visualization:* Kamila Malesa. *Writing—original draft:* Ewa Tramś and Kamila Malesa. *Writing—review and editing:* Ewa Tramś, Kamila Malesa, Krzysztof Kuliński and Rafał Kamiński. All authors have approved the manuscript and agree with its submission to Knee Surgery, Sports Traumatology, Arthroscopy.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ETHICS STATEMENT

The authors have nothing to report.

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SUPPORTING INFORMATION

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